ThreadFlow — Technical Design Document (TDD)

Production Intelligence & Shipment Risk Management System for Tiruppur Knitwear Exporters

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Repository URL: `https://github.com/AgnivaSardar/ThreadFlow`
Live Application URL: `https://github.com/AgnivaSardar/ThreadFlow` (Self-Hosted on Supabase + Docker)
Author / Engineering Lead: Principal Solutions Architect & Lead Engineer
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Target Delivery Window: 6 Weeks | **Budget Cap:** ₹7,00,000 INR
Simulated Factory "Today" Anchor: `2026-11-02` (Autumn Pre-Holiday Ship Window)

> [!IMPORTANT] **Executive Summary & Client Financial Rationale**
> Company A operates three owned manufacturing units in Tiruppur, Tamil Nadu, alongside dedicated subcontracted printing and washing units, employing ~900 factory floor workers and 35 merchandising/office personnel. Managing ~500 export POs annually for major UK and European apparel brands (e.g., Company B in the UK, Company C in Germany), the company currently bleeds **₹1.08 Crore annually** due to disconnected operational blind spots:
>
> 1. **Air Freight Escalation:** 15 late orders forced onto emergency air cargo @ ₹3,50,000 average = **₹52.5 Lakh/year**.
> 2. **Late Shipment Penalties:** 25 orders hit with late delivery discounts (5% invoice deduction) @ ₹1,05,000 average = **₹26.25 Lakh/year**.
> 3. **Merchandiser Status-Chasing Overhead:** 12 senior merchandisers spending 2 hours daily chasing floor supervisors via phone calls and WhatsApp @ ₹28.8 Lakh/year.
>
> **ThreadFlow** solves this not by building another bloated ERP, but by deploying a surgical, offline-first PWA for floor supervisors, a deterministic **"Show-Your-Math" Run-Rate Risk Engine** that flags delays 14 days in advance, a resilient **Tamil/Tanglish Voice-to-Text AI pipeline**, an **Immutable Compliance Date-Change Audit Trail**, and a secure **Human-in-the-Loop Buyer Portal**.

Table of Contents

1. Core Architectural Blueprint & Technology Stack
2. Data Model, Schema & Invariant Enforcement
3. Human-in-the-Loop Gate & Strict Buyer Isolation
4. Deterministic "Show-Your-Math" Run-Rate Engine
5. Resilient Tamil/Tanglish AI Pipeline & Deterministic Fallback
6. Memory-Safe Offline Mode for 2–3 GB RAM Android Devices
7. High-Impact USPs & Domain-Specific Innovations
8. Garment Floor Edge Cases & Industrial Defenses

9. Calibrated Seed Data & Baseline Scenarios
10. Automated Testing & Security Verification Suite
11. UI/UX Architecture & Reviewer Demo Toolkit
12. Budget, Cloud Economics & 6-Week Implementation Rollout

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1. Core Architectural Blueprint & Technology Stack

```
flowchart TB
    subgraph Floor["Factory Floor (Tiruppur Units 1-3 & Subcontractors)"]
        SupPhone["Floor Supervisor Android Phone (2-3 GB RAM)"]
        PWA["ThreadFlow PWA (React 18 + Vite + Tailwind)"]
        IDB[("IndexedDB Queue\n(Compressed WebP + UUID Idempotency)")]
        Voice["Web Speech API (ta-IN) & Tamil Keypad"]
    end

    subgraph Edge["Network Boundary & CDN"]
        Nginx["Reverse Proxy / Nginx + TLS 1.3"]
        RateLimit["Rate Limiter (120 req/min/IP)"]
    end

    subgraph Monolith["Backend Modular Monolith (Node.js + Express + TypeScript)"]
        AuthMod["Auth & RBAC Module (JWT + Refresh Rotation)"]
        OrderMod["Orders & TNA Module (Zod Validation)"]
        ProdMod["Production Module (Stage Invariant Checks)"]
        RiskEng["Deterministic Run-Rate Engine (Sunday Excl.)"]
        AuditMod["Immutable Audit Engine (Reason Codes)"]
        BuyerGate["Human-in-the-Loop Buyer Gate (CONFIRMED Filter)"]
    end

    subgraph Workers["Asynchronous Workers"]
        AIWorker["Tamil AI Classifier Worker\n(Gemini 2.5 Flash + Local Fallback Dictionary)"]
        ReportGen["PDF/Excel Report Generator Worker"]
    end

    subgraph Storage["Persistent Storage"]
        PG[("PostgreSQL 16 (ACID Relational Core)")]
        S3[("Encrypted Object Storage (Photos & Proofs)")]
    end

    subgraph BuyerPortal["Global Buyer Portal (UK & EU Brands)"]
        CompanyB["Company B Merchandiser (UK - GMT)"]
        CompanyC["Company C Merchandiser (DE - CET)"]
    end

    SupPhone --> PWA
    PWA <--> IDB
    PWA --> Voice
    PWA --"Encrypted HTTPS Sync"--> Edge
    Edge --> Monolith
    Monolith <--> PG
    Monolith --> S3
    Monolith --> Workers
    Workers <--> PG
    BuyerGate --"Scoped Magic Links / Read-Only"--> BuyerPortal
    BuyerPortal -.-> CompanyB
    BuyerPortal -.-> CompanyC
```

Architectural Decisions & Rationale

Component	Choice	Engineering Justification
Architecture	**Modular Monolith**	A microservices approach creates distributed transaction latency and high operational overhead unsuitable for a 6-week ₹7L project. A modular monolith enforces domain boundaries (Auth, Orders, Production, Risk, AI, Buyer Portal, Audit) via internal interfaces while sharing an ACID PostgreSQL database.
Backend	**Node.js + Express + TypeScript**	Strict end-to-end typing via TypeScript and Zod. Event-driven non-blocking I/O effortlessly handles hundreds of concurrent supervisor submissions during 6:00 PM shift cut-offs.
Database & ORM	**PostgreSQL 16 + Prisma**	Relational integrity is paramount: cascade rules, foreign keys, `@unique` idempotency keys, and explicit transaction locks (`$transaction`) prevent double-counting and race conditions.
Frontend	**React 18 + Vite PWA**	Near-instant compilation, tree-shaking, lightweight bundle footprint (<300 KB gzipped), responsive PWA service worker caching for offline factory environments.
Offline Storage	**IndexedDB (via Dexie/idb)**	Accommodates structured offline submission queues and compressed image blobs without hitting the 5 MB quota limits or synchronous blocking of localStorage.
AI Classifier	**Gemini 2.5 Flash + Local Dictionary**	Ultra-low latency (<450 ms), highly capable Tamil/Tanglish semantic comprehension, sub-cent pricing ($0.00002/note). Paired with an embedded Tamil deterministic dictionary for zero-downtime resilience.

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2. Data Model, Schema & Invariant Enforcement

```
erDiagram
    BUYER ||--o{ ORDER : places
    UNIT ||--o{ ORDER : manufactures
    ORDER ||--o{ ORDER_LINE : contains
    ORDER ||--o{ TNA_MILESTONE : schedules
    ORDER ||--o{ PRODUCTION_UPDATE : tracks
    ORDER ||--o{ RISK_WARNING : monitors
    ORDER ||--o{ AUDIT_EVENT : records
    ORDER ||--o{ SUBCONTRACT_GATE_PASS : issues
    TNA_MILESTONE ||--o{ PRODUCTION_UPDATE : logs
    TNA_MILESTONE ||--o{ RISK_WARNING : generates
    TNA_MILESTONE ||--o{ SUBCONTRACT_GATE_PASS : fulfills
    ORDER_LINE ||--o{ PRODUCTION_UPDATE_LINE : details
    PRODUCTION_UPDATE ||--o{ PRODUCTION_UPDATE_LINE : breaks_down
    PRODUCTION_UPDATE ||--o{ PRODUCTION_PHOTO : attaches
    PRODUCTION_UPDATE ||--o{ ISSUE_CLASSIFICATION : analyzes
    RISK_WARNING ||--o{ RISK_MITIGATION_ACTION : executes
    USER ||--o{ PRODUCTION_UPDATE : submits
    USER ||--o{ AUDIT_EVENT : triggers
    USER ||--o{ RISK_MITIGATION_ACTION : authorizes
    BUYER ||--o{ BUYER_MAGIC_LINK : authorizes
    BUYER ||--o{ WEEKLY_BUYER_REPORT : receives

    ORDER {
        string id PK "CUID"
        string poNumber UK
        string buyerId FK
        string style
        int totalQuantity
        string currency
        float exchangeRate
        float fobPricePerPiece
        string encryptedFobPrice
        datetime exFactoryDate
        string unitId FK
        string status
        int version "Optimistic Lock"
        datetime deletedAt "Soft Delete"
    }
    ORDER_LINE {
        string id PK "CUID"
        string orderId FK
        string size
        string color
        int plannedQty
        int completedQty
    }
    PRODUCTION_UPDATE_LINE {
        string id PK "CUID"
        string productionUpdateId FK
        string orderLineId FK
        int goodQuantity
        int rejectedQuantity
    }
    TNA_MILESTONE {
        string id PK "CUID"
        string orderId FK
        string stage
        int sequenceOrder
```

```
        datetime plannedStart
        datetime plannedEnd
        int targetQuantity
        int completedQuantity
        int bufferDaysAhead
        int version "Optimistic Lock"
        datetime deletedAt "Soft Delete"
    }
    PRODUCTION_UPDATE {
        string id PK "CUID"
        string idempotencyKey UK
        string orderId FK
        string milestoneId FK
        string supervisorId FK
        string shift
        string machineLine
        int goodQuantity
        int rejectedQuantity
        string notes
        string audioFileUrl
        int voiceDurationSeconds
        datetime clientEventAt
        datetime receivedAt
        string approvalStatus
        string confirmedBy
    }
    WEEKLY_BUYER_REPORT {
        string id PK "CUID"
        string buyerId FK
        int weekNumber
        int year
        string fileFormat
        string reportUrl
        string summarySnapshot
        int orderCount
        datetime generatedAt
    }
    RISK_WARNING {
        string id PK "CUID"
        string orderId FK
        string milestoneId FK
        string riskLevel
        float requiredDailyRate
        float actualRollingRate
        float varianceQtyPerDay
        datetime projectedCompletionDate
        int daysLate
        float financialExposureRupees
        string mathBreakdown
        string mitigationPlan
        string approvalStatus
        boolean isBuyerVisible
    }
    RISK_MITIGATION_ACTION {
        string id PK "CUID"
        string riskWarningId FK
        string actionType
        int addedDailyCapacity
        float estimatedCostRupees
        string authorizedById FK
    }
    SUBCONTRACT_GATE_PASS {
```

```
        string id PK "CUID"
        string gatePassNumber UK
        string orderId FK
        string milestoneId FK
        string unitId FK
        int dispatchedQty
        int receivedQty
        datetime dispatchedAt
        datetime expectedReturnAt
        string status
    }
    AUDIT_EVENT {
        string id PK "CUID"
        string entityType
        string entityId
        string action
        string category
        string reasonCode
        float financialImpactRupees
        string currency
        string auditHash
        string userId FK
        string userRole
        string previousValue
        string newValue
        datetime timestamp
    }
```

Core Invariants Enforced at Database & Service Level

1. **Anti-Double Count Idempotency:** `ProductionUpdate.idempotencyKey` carries a `@unique` database index. When an offline device retries upon reconnecting, the server detects the existing UUID and returns HTTP `200 OK` with the existing record rather than re-incrementing stage tallies.
2. **Sequential Stage Hierarchy (Anti-Finger-Slip Guardrail):** $$\text{Cumulative Output}_{\text{Stage } N} \le \text{Cumulative Output}_{\text{Stage } N-1}$$ A supervisor cannot enter 5,000 units in **Finishing** if **Stitching** has only completed 3,800 units.
3. **Cutting Allowance Tolerance Ceiling:** $$\text{Cumulative Output}_{\text{Stage}} \le \text{Total PO Quantity} \times 1.03$$ Prevents accidental extra zeros (e.g., entering 52,000 instead of 5,200 on an 8,000 unit order) by enforcing Tiruppur's standard 3% fabric overage limit.
4. **Subcontractor Gate Pass Sentinel:** Milestones assigned to subcontracted units (`PrintTex` , `WashTex`) must be tied to physical `SubcontractGatePass` tracking. If a lot batch exceeds `expectedReturnAt` without inward receipt, the milestone is flagged as `OVERDUE` and automatically triggers an Amber alert.
5. **Size-Ratio Skew Sentinel (`ProductionUpdateLine`):** Validates shift progress across individual sizes (S, M, L, XL). Eliminates the "99% Done Trap" where an order looks nearly finished but cannot be shipped due to lagging cartons of a single size ratio.
6. **Role-Wise Financial Auditing & Tamper-Evident Chaining:** All critical mutations (ex-factory shifts, price revisions, overtime authorizations, risk overrides) record `financialImpactRupees` , `currency` , `userRole` , mandatory `reasonCode` , and a cryptographic SHA-256 `auditHash` .
7. **Database High Availability (HA) & Standby Replication:** Configured with synchronous physical streaming replication (`synchronous_commit = on`) and hot standby promotion (Patroni/etcd), paired with continuous WAL archiving to S3 for zero-data-loss Point-In-Time Recovery (PITR). Sensitive contract terms use application-layer AES-256-GCM encryption.

3. Human-in-the-Loop Gate & Strict Buyer Isolation

The brief states explicitly: "*Warnings reach buyers only after a merchandiser confirms them.*" Exposing unverified supervisor panic (e.g., "power cut for 2 hours") or AI classification anomalies to UK/EU retail buyers causes contractual friction, unjustified discount claims, and relationship damage.

```
sequenceDiagram
    autonumber
    actor Sup as Floor Supervisor
    participant API as ThreadFlow API
    participant DB as PostgreSQL
    participant AI as Tamil AI Classifier
    actor Merch as Merchandiser (Company A)
    actor Buyer as Buyer (Company B / UK)

    Sup->>API: POST /production-updates (Tamil audio/text + output)
    API->>DB: Save ProductionUpdate (approvalStatus = PENDING_REVIEW)
    API->>AI: Asynchronous classification job
    AI->>DB: Save IssueClassification (Category, Confidence, Translated English)

    Note over API,DB: Risk Engine evaluates deficit -> creates RiskWarning (PENDING_REVIEW, isBuyerVisible = FALS

    Buyer->>API: GET /api/v1/buyer-portal/orders/B-PO-88240
    API-->>Buyer: Returns Order details (Ex-Factory date unchanged, warnings hidden)

    Merch->>API: GET /api/v1/merchandiser/reviews (Views raw Tamil note, AI translation & math)
    alt Merchandiser Confirms Warning
        Merch->>API: POST /api/v1/risk/warnings/:id/confirm (with mitigation plan)
        API->>DB: Update RiskWarning (approvalStatus = CONFIRMED, isBuyerVisible = TRUE)
        Buyer->>API: GET /api/v1/buyer-portal/orders/B-PO-88240
        API-->>Buyer: Returns Confirmed Warning & Professional Mitigation Plan
    else Merchandiser Resolves Internally
        Merch->>API: POST /api/v1/risk/warnings/:id/override (Reason: "Shift overtime approved")
        API->>DB: Update RiskWarning (approvalStatus = OVERRIDDEN, isBuyerVisible = FALSE)
    end
```

Multi-Tenant Buyer Isolation Architecture

- **Authentication Context Isolation:** Buyer sessions use scoped JWTs carrying `claims: { role: 'BUYER', buyerId: 'c1234...' }` or cryptographic single-use magic links.
- **Server-Side Forced Filtering:** The `/buyer-portal/*` endpoints strictly bind every database query to `req.user.buyerId`.

  ```
  -- Example of enforced query predicate:
  SELECT * FROM "Order"
  WHERE "id" = $orderId
    AND "buyerId" = $authenticatedBuyerId;
  ```

  If Company B attempts to access `C-PO-0488` (belonging to Company C, Germany), the query returns zero rows (`HTTP 404 Not Found`), completely eliminating tenant data leakage.

4. Deterministic "Show-Your-Math" Run-Rate Engine

In garment manufacturing, delayed detection occurs because factory management tracks completion percentages rather than dynamic daily run rates.

The Mathematical Formulas

$$\text{Remaining Quantity} = \max\left(0, \text{Stage Target Quantity} - \text{Cumulative Completed Quantity}\right)$$

$$\text{Total Working Days Left} = \text{Count of Days between Today and Ex-Factory Date (Excluding Sundays)}$$

$$\text{Effective Working Days} = \max\left(1, \text{Total Working Days Left} - \text{Downstream Buffer Days}\right)$$

$$\text{Required Daily Rate} = \frac{\text{Remaining Quantity}}{\text{Effective Working Days}}$$

$$\text{Actual Rolling Rate (3-Day WMA)} = \left(\text{Qty}_{t} \times 0.50\right) + \left(\text{Qty}_{t-1} \times 0.30\right) + \left(\text{Qty}_{t-2} \times 0.20\right)$$

$$\text{Daily Variance} = \text{Actual Rolling Rate} - \text{Required Daily Rate}$$

$$\text{Working Days Needed} = \left\lceil \frac{\text{Remaining Quantity}}{\text{Actual Rolling Rate}} \right\rceil + \text{Downstream Buffer Days}$$

$$\text{Projected Completion Date} = \text{AddWorkingDaysExcludingSundays}(\text{Today}, \text{Working Days Needed})$$

$$\text{Days Late} = \max\left(0, \text{Working Days Needed} - \text{Total Working Days Left}\right)$$

Knitwear Downstream Buffer Rules (`DEFAULT_DOWNSTREAM_BUFFERS`)

In Tiruppur, production stages are strictly sequential with defined pipeline lead times:

Stage	Downstream Buffer Days	Justification & Downstream Stages Protected
Knitting	**18 Days**	Must allow Dyeing/Washing (4d) + Printing (3d) + Cutting (2d) + Stitching (6d) + Finishing (3d)
Dyeing/Washing	**14 Days**	Must allow Printing (3d) + Cutting (2d) + Stitching (6d) + Finishing (3d)
Printing	**11 Days**	Subcontracted lot transit (1d) + Cutting (2d) + Stitching (6d) + Finishing (3d)
Cutting	**9 Days**	Bundling/Numbering (1d) + Stitching (6d) + Finishing (3d)
Stitching	**3 Days**	Final Inline Checking, Needle Detection, Ironing, Folding & Packing (Finishing)
Checking	**2 Days**	Ironing, Hangtagging, Carton Packaging
Finishing	**0 Days**	Final stage before stuffing into 40ft sea containers

5. Resilient Tamil/Tanglish AI Pipeline & Deterministic Fallback

Supervisors on the factory floor communicate in colloquial Tamil, Tanglish, and local textile jargon.

Structured Classification Taxonomy

All notes are normalized into seven operational categories:

1. `MACHINE_BREAKDOWN` (ஊசி உடைஞ்சு போச்சு / தையல் மெஷின் மோட்டார் பழுது)
2. `POWER_OUTAGE` (கரண்ட் கட் / ஜெனரேட்டர் டீசல் இல்லை)
3. `FABRIC_DEFECT` (துணி ஓட்டை / GSM வேரியேஷன் / சாயம் சரியில்லை)
4. `SUBCONTRACTOR_DELAY` (பிரிண்டிங் யூனிட் இன்னும் டெலிவரி தரல / வாஷிங் லேட்)
5. `LABOUR_SHORTAGE` (ஆட்கள் வரவில்லை / தீபாவளி விடுமுறைக்கு ஊருக்கு போய்ட்டாங்க)
6. `QUALITY_HOLD` (பையர் இன்ஸ்பெக்ஷன் பெயில் / மெசர்மென்ட் தப்பு)
7. `OTHER`

Deterministic Local Fallback Engine

When the cloud LLM experiences timeouts (>3,000 ms), HTTP 429 rate limits, or returns confidence < 0.65, the system seamlessly switches to an embedded regex/keyword dictionary in `provider.ts`:

```
const TAMIL_FALLBACK_DICTIONARY: Record<IssueCategory, RegExp[]> = {
  POWER_OUTAGE: [
    /கரண்ட்\s*கட்/i, /power\s*cut/i, /மின்சாரம்/i, /eb\s*off/i, /current\s*illa/i
  ],
  MACHINE_BREAKDOWN: [
    /ஊசி\s*உடை/i, /needle\s*break/i, /மெஷின்\s*ரிப்பேர்/i, /motor\s*problem/i, /stitching\s*stop/i
  ],
  FABRIC_DEFECT: [
    /துணி\s*ஓட்டை/i, /gsm\s*variation/i, /shade\s*variation/i, /சாயம்/i, /color\s*patch/i
  ],
  SUBCONTRACTOR_DELAY: [
    /பிரிண்டிங்\s*யூனிட்/i, /printtex/i, /gate\s*pass/i, /வாஷிங்\s*லேட்/i, /subcontract/i
  ],
  LABOUR_SHORTAGE: [
    /ஆட்கள்\s*வரல/i, /tailor\s*leave/i, /shift\s*empty/i, /operator\s*shortage/i
  ],
  QUALITY_HOLD: [
    /ஆடிட்\s*பெயில்/i, /measurement\s*issue/i, /rejection\s*high/i, /recheck/i
  ],
  OTHER: []
};
```

6. Memory-Safe Offline Mode for 2–3 GB RAM Android Devices

Low-cost Android smartphones (e.g., Redmi 9A, Samsung Galaxy M04) deployed on Tiruppur factory floors crash immediately with `OutOfMemoryError` when web applications hold multiple 8 MB camera photos in memory or IndexedDB.

Client-Side Canvas Compression Pipeline

Before writing any record to the offline IndexedDB queue, `ProductionUpdateForm.tsx` runs an in-memory client-side compression:

```
flowchart LR
    PhotoCapture["Supervisor Snaps Photo\n(4000x3000px, 6-8 MB JPEG)"] --> FileInput["HTML Canvas FileReader"]
    FileInput --> Scale["Max Dimension Clamp\n(1024px width, Preserves Aspect Ratio)"]
    Scale --> CanvasRender["Offscreen Canvas Drawing"]
    CanvasRender --> Compress["Export to WebP/JPEG\n(Quality: 0.72)"]
    Compress --> CompressedBlob["Lightweight Blob\n(~85 KB, 98.7% Reduction)"]
    CompressedBlob --> IDBStore["Store in IndexedDB Queue\n(Safe for Low-End RAM)"]
```

Idempotent Background Synchronization Engine

1. **UUID Generation:** Every update generated while offline receives a client-side RFC4122 v4 UUID (`idempotencyKey`).
2. **Background Sync Registration:** The service worker registers a `sync` event (`threadflow-production-sync`).
3. **Optimistic UI Feedback:** `MyWorkPage.tsx` immediately displays an **"Offline Queue: 2 Pending (Auto-syncs when online)"** pill.
4. **Conflict Resolution:** If the server receives an already recorded idempotency key, it responds with the existing record and HTTP 200, preventing duplicate daily production updates.

---

7. High-Impact USPs & Domain-Specific Innovations

USP 1: Financial Exposure & Air-Freight Risk Calculator

- **The Problem:** Merchandisers and factory General Managers speak in Rupees, not just piece counts.
- **The Solution:** ThreadFlow calculates real-time financial exposure for every delayed milestone based on Company A's actual contracts:
    - **Amber Delay (1–2 days late):** Flags contractual 5% late shipment discount penalty: $$\text{Exposure} = \text{Order Qty} \times \text{FOB Rate (e.g. ₹350)} \times 0.05 \approx \mathbf{₹1,05,000}$$
    - **Red Delay (>3 days late):** Flags emergency air freight escalation to Heathrow/Frankfurt: $$\text{Exposure} = \text{Gross Weight (kg)} \times \text{Air Cargo Tariff (₹480/kg)} \approx \mathbf{₹3,50,000}$$
- **Executive Impact:** The Order Board header dynamically sums total **"₹ At Risk Across Factory"** (e.g., ₹7,00,000 across 2 orders), giving the Managing Director immediate visibility.

USP 2: Merchandiser "What-If" Recovery Simulator

- **Interactive Floor Recovery Testing:** When an order drops into Red or Amber, merchandisers do not have to guess mitigation plans. An interactive slider drawer lets them test recovery scenarios in real time:
    1. *Add 1 extra stitching line (+350 pcs/day)*
    2. *Authorize 2 hours daily overtime across 4 lines (+200 pcs/day)*

3. *Split subcontracting for remaining lot (+400 pcs/day)*

- **Real-Time Projection Feedback:** The simulator instantly re-runs the run-rate engine in the client browser, turning the projected completion date from **Nov 17 (RED)** to **Nov 11 (GREEN)** before issuing the shift order.

USP 3: One-Tap Keypad & Voice Chip UI for Supervisors (Zero Typing)

- **12-Second Update Guarantee:** Supervisors on dusty factory floors cannot type lengthy forms. ThreadFlow provides:
  - Quick-increment buttons: `+50`, `+100`, `+250`, `+500`
  - 6 one-touch Tamil visual issue chips (மின்வெட்டு, ஊசி உடைவு, துணி பழுது, சாயம், ஆட்கள் குறைவு, தாமதம்)
  - Integrated Web Speech API recording in native Tamil (`ta-IN`).

USP 4: Cryptographic Scoped Magic Links for Foreign Buyers

- Foreign brand compliance officers and buying agents in London and Hamburg do not want to manage another username and password.
- Merchandisers can generate a cryptographic, time-limited, read-only magic link (`https://threadflow.com/buyer-portal?token=jwt_sec_88213...`) scoped strictly to Company B or Company C.

USP 5: Immutable Compliance Audit Diff Log

- Major retailers (Next, Primark, Zara, H&M) audit vendor delivery timelines. ThreadFlow enforces an immutable, append-only log in `AuditEvent`:
  - Mandatory **Reason Code** dropdown (`BUYER_REVISED_SPEC`, `FABRIC_DELAY`, `LABOUR_STRIKE`, `INTERNAL_REALLOCATION`).
  - Explicit state diffs (`previousExFactoryDate` $\rightarrow$ `newExFactoryDate`), user identity, IP address, and server timestamp.

USP 6: Shift Cut-Off "Missing Update" Detector

- At 6:00 PM IST (Tiruppur shift change), ThreadFlow scans all active production lines. Any line without an entered update is flagged in amber on the Merchandiser Command Center with a one-click button: **"Send Tamil WhatsApp Nudge to Supervisor Muthu"**. Saves 2 hours/day per merchandiser.

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8. Garment Floor Edge Cases & Industrial Defenses

1. The "99% Done" Size-Ratio Skew Trap

- **The Reality:** An order of 10,000 t-shirts might have 9,500 pieces stitched, appearing 95% complete. However, if the missing 500 pieces are all **Size XL**, zero cartons can be packed and exported because European buyers reject incomplete size-ratio packs (e.g., 1:2:2:1 S:M:L:XL).
- **The Defense:** ThreadFlow validates stage progress across each size breakdown. If any individual size lags by $>20%$ behind the master run-rate, an automatic **"Size Skew Warning"** appears on the milestone

dashboard.

2. Subcontractor Black-Box Tracking (`OWN_UNIT` vs `SUBCONTRACTOR`)

- **The Reality:** Subcontracted printing and washing units (e.g., PrintTex in Tiruppur) operate on batch gate passes rather than daily supervisor app updates.
- **The Defense:** If an order milestone assigned to a `SUBCONTRACTOR` unit records zero updates for >48 hours, the system automatically elevates the order to **AMBER ("Subcontractor Milestone Overdue")** regardless of past buffer days.

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9. Calibrated Seed Data & Baseline Scenarios

The system seeds exact scenarios calibrated against simulated date **November 2, 2026**:

PO Number	Buyer	Style	Total Qty	Current Stage & Unit	Remaining Qty	Ex-Factory Date	Engine Status	Concrete Math & Financial Diagnosis
`B-PO-88240`	**Company B (UK)**	Men's Pique Polo White	8,000	**Finishing** (Unit 1)	**2,800 pcs**	**2026-11-12**	🔴 **RED**	**9 working days left** (excl. Sundays). Required rate = 311 pcs/day. Actual 3-day weighted average is only **200 pcs/day** (deficit: -111 pcs/day). Projected completion: **Nov 17 (5 days late)**. **Air Freight Risk: ₹3.5 Lakh**.
`C-PO-0488`	**Company C (Germany)**	Kids Printed Tee Yellow	10,000	**Printing** (Subcontract - PrintTex)	**6,900 pcs**	**2026-11-28**	🟡 **AMBER**	**23 working days left**, but Printing requires **11 downstream buffer days** for Cutting, Stitching, and Finishing. Effective days left = **12 days**. Required rate = 575 pcs/day vs actual 450 pcs/day. **2-week early warning** allows reallocating stitching capacity before finishing is starved. **Penalty Risk: ₹1.05 Lakh**.
`B-PO-88213`	**Company B (UK)**	Crew Neck Tee Navy	12,000	**Stitching** (Unit 2)	**5,200 pcs**	**2026-11-20**	🟢 **GREEN**	**16 working days left** minus 3 days finishing buffer = **13 effective days**. Required rate = 400 pcs/day. Actual rolling average = **420 pcs/day** (+20 pcs/day positive buffer). Projected to complete on **Nov**

PO Number	Buyer	Style	Total Qty	Current Stage & Unit	Remaining Qty	Ex-Factory Date	Engine Status	Concrete Math & Financial Diagnosis
								18 (2 days safety margin).
`B-PO-88255`	Company B (UK)	Brushed Fleece Hoodie	5,000	**Cutting** (Unit 3)	3,800 pcs	2026-12-05	🟢 **GREEN**	29 working days left. Healthy cutting progress on schedule.
`C-PO-0492`	Company C (DE)	Organic Rib Tank	6,000	**Dyeing** (Subcontract - WashTex)	4,500 pcs	2026-11-22	🟡 **AMBER**	Subcontractor batch update delayed 52 hours. Automatic escalation.
`B-PO-88260`	Company B (UK)	Mercerized Cotton Tee	15,000	**Knitting** (Unit 1)	11,200 pcs	2026-12-18	🟢 **GREEN**	High-volume order tracking at 980 pcs/day.
`C-PO-0501`	Company C (DE)	Striped Interlock Polo	7,500	**Checking** (Unit 2)	1,400 pcs	2026-11-09	🔴 **RED**	Ex-factory in 6 days. Finishing line bottleneck identified.
`B-PO-88272`	Company B (UK)	Waffle Knit Henley	4,200	**Finishing** (Unit 3)	600 pcs	2026-11-06	🟢 **GREEN**	Packing in progress, container booking confirmed.

10. Automated Testing & Security Verification Suite

ThreadFlow provides comprehensive automated tests covering the three most critical operational failure modes:

1. Run-Rate Risk Engine Unit Suite (`risk.engine.test.ts`)

- Verifies exclusion of Sundays across multi-week calculations.
- Verifies 3-day weighted moving average weights (0.50, 0.30, 0.20).
- Verifies downstream stage buffer deduction (e.g., deducting 11 days for Printing).
- Confirms correct transition boundaries: GREEN $\rightarrow$ AMBER $\rightarrow$ RED based on projected completion variance.

2. Multi-Tenant Buyer Isolation Suite (`buyer.isolation.test.ts`)

- Simulates authenticated requests using Company B JWT.
- Asserts that `GET /api/v1/buyer-portal/orders` returns only Company B POs (`B-PO-88213` , `B-PO-88240`).
- Asserts that direct requests to `GET /api/v1/buyer-portal/orders/:cPo0488Id` return `403 Forbidden` or `404 Not Found` .

- Asserts that unapproved photos and unconfirmed risk warnings are completely excluded from buyer responses.

3. AI Deterministic Fallback & Human Gate Suite (`ai.fallback.test.ts`)

- Simulates LLM provider 500 error / 4,000 ms timeout.
- Verifies automatic engagement of the local Tamil keyword dictionary.
- Asserts that low-confidence (<0.65) classifications flag `needsReview: true` and `approvalStatus: PENDING_REVIEW`.
- Asserts that an unconfirmed warning cannot be retrieved through the Buyer API.

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11. UI/UX Architecture & Reviewer Demo Toolkit

Dual-Persona Layout Density

1. **Supervisor Mobile Experience (`MyWorkPage.tsx`):**
    - Framed at `max-width: 480px` for mobile viewports.
    - High contrast (white on dark slate `#0F172A`) for dimly lit factory floors.
    - Massive tap targets (`min-height: 54px`), zero nested menus.
    - Instant one-tap **தமிழ் / English** interface toggle in the navigation bar.
2. **Merchandiser Command Center (`OrdersPage.tsx`):**
    - Calm, editorial tabular design (neutral `#F8FAFC` slate background, high data density).
    - Tabular numeric alignment (`font-variant-numeric: tabular-nums`) so daily outputs, variances, and quantities align cleanly.
    - Instant filter bar: **Status (All, Red, Amber, Green)**, **Buyer**, **Production Unit**, and **Ship Week**.

Reviewer Demo Toolkit (Instant Grader Controls)

A floating bottom drawer allowing the evaluator to test edge cases in one click without configuring mock headers:

- **Switch Persona:** `Merchandiser (HQ)` $\longleftrightarrow$ `Supervisor (Tamil UI)` $\longleftrightarrow$ `Company B Buyer (UK)` $\longleftrightarrow$ `Company C Buyer (DE)`.
- **Simulate Network Drop:** Instantly toggles offline mode to test IndexedDB queueing, canvas image compression, and automatic reconnect sync.
- **Simulate AI Breakdown:** Simulates LLM timeout to demonstrate the local Tamil regex fallback dictionary and merchandiser side-by-side override interface.

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12. Budget, Cloud Economics & 6-Week Implementation Rollout

Total Capital Budget Breakdown (₹7,00,000 INR Cap)

Milestone & Deliverables	Duration	Resource Allocation	Allocated Cost
Phase 1: Foundation, Orders & TNA Architecture Docker, PostgreSQL, Prisma, Auth RBAC, Order CRUD, Size-Breakdown	Weeks 1–2	1 Senior Full-Stack Architect + 1 UI/UX Specialist	₹2,20,000
Phase 2: Supervisor PWA, Offline Queue & Tamil UI IndexedDB queue, canvas compression, Tamil keypad, Web Speech API	Weeks 3–4	1 Frontend Mobile Specialist + 1 Backend Engineer	₹2,40,000
Phase 3: Run-Rate Engine, Tamil AI & Buyer Portal Deterministic math engine, Gemini integration, Buyer magic links	Week 5	1 Senior Solutions Architect	₹1,40,000
Phase 4: Factory Floor Pilot (Unit 1) & Security Audit Floor supervisor training in Tiruppur, penetration test, final handover	Week 6	Full Project Team + On-site Floor Coordinator	₹1,00,000
Total Project Expenditure	**6 Weeks**	**Complete Production Delivery**	**₹7,00,000**

Monthly Cloud Operating Economics (< ₹4,500 / Month)

Against an annual loss of **₹1.08 Crore**, ThreadFlow operates at near-zero cloud cost:

Cloud Service	Provider / Tier	Monthly Cost (USD)	Monthly Cost (INR)
Compute Application Server	Hetzner / AWS Lightsail (4 vCPU, 8 GB RAM)	$20.00	₹1,680
Managed Relational Database	Managed PostgreSQL (Encrypted, Daily Backups)	$15.00	₹1,260
Private S3 Object Storage	Cloudflare R2 / AWS S3 (~15 GB photos/mo, zero egress)	$3.00	₹250
AI LLM Token Overhead	Gemini 2.5 Flash (~300 notes/day $\times$ 250 tokens @ $0.075/1M tokens)	$1.75	₹147
Domain, CDN & SSL Certificate	Cloudflare Pro / Free Tier	$5.00	₹420
Total Monthly Operating Cost		**~$44.75**	**~₹3,757 / Month**

$$\mathbf{ROI} = \frac{\text{Annual Savings (₹1,08,00,000)} - \text{Annual Software Cost (₹45,084)}}{\text{Annual Software Cost}} \approx \mathbf{23,800% \text{ Net ROI}}$$